What Could This Be? A Comparative Study of Standard and Non-Standard Dark Matter Theories

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December 2025

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1 Introduction

In 1933, Swiss-American astronomer Fritz Zwicky happened upon the observation that the Coma Cluster hadn't ripped itself apart yet, even though its constituent galaxies seemed to be moving so quickly they should have launched themselves away into deep space. Like any curious scientist does, he chose to investigate this paradox, and found that the theoretical mass of the cluster was one hundred times greater than what could be seen![1] Concluding that there was definitely a gravitational anomaly at work, he chose to interpret it as invisible mass that only interacts with what we can see via gravity, christening this hypothetical object "dark matter" and opening a Pandora's box of mysteries for at

least a century to come. Since then, scientists throughout history and the world have worked together to solve the dark matter enigma, through decades of theories and detectors. But its identity remains elusive still, leaving one big question up in the air which this essay will examine in depth... what could it be?

2 Why Dark Matter?

To start this treatise on dark matter properly, a reasonable first question is, "why are we so sure that the cause of these gravitational anomalies is invisible flying stuff?" This is a very fair question and should be approached with careful scrutiny. Until 1970, most scientists shared thus opinion. Then Albert Bosma in conjunction with Morton Roberts and Ken Freeman chose to observe the speeds at which matter in a galaxy rotates around its center, plotting their prediction against what was actually observed, showing very anomalous behaviour as outlined in Figure 1!

So, there definitely is a widespread gravitational anomaly. But what is it really? It could just be that we got gravity wrong, right? Observing the Bullet Cluster very quickly disproves this, as one careful look at this galaxy merger and it's readily apparent how its center of mass is offset from its center of gravitational lensing. This effect is caused by high mass density, which means that the bulk of the Cluster's mass is invisible, which modified gravity cannot explain. Okay, so, it's not modified gravity. What is it then?

The cosmic microwave background (CMB), a near-ideal blackbody made of the last photons emitted during the universe's inflationary era, can help us here. It notably contains anisotropies, with particular attention paid to the highest deviations from baseline (the "acoustic peaks"). The maths for this process is done in Fourier space, starting with a decomposition of the temperature field's monopole via Fourier transformation and neglecting gravity in our flat universe[2]:

$$\Theta_{l=0,m=0}(\vec{x}) = \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}} \Theta(\vec{k}) \tag{1}$$

When evaluated by plugging a modified Euler equation into the differentiated continuity equation, this leads to an equation for Fourier space perturbations in the temperature field:

$$\ddot{\Theta}(\vec{k}) + c_s^2 k^2 \Theta = 0 \tag{2}$$

By recombination, the temperature distribution from this oscillation is mapped as:

$$\Theta(\eta_*) = \Theta(0)\cos(k s_*) \tag{3}$$

After applying general relativity and considering gravity and causality, the post-recombination temperature distribution looks like:

$$\Theta = -\frac{\delta a}{a} = -\frac{2}{3}\left(1 + \frac{p}{\rho}\right)^{-1} \frac{\delta t}{t} \tag{4}$$

After obtaining the acoustic peaks of the CMB, it can be matched up against predictions assuming only baryonic matter, modified gravity, or dark matter; only the last of these is in agreement as shown in Figure. Additionally, a visual map of the CMB has its anisotropies roughly correlate with the present day distributions of galaxies, implying that galaxies and dark matter cluster around each other which would explain the previous observations.[3]

Therefore, we have established the almost certain existence of dark matter as a previously unknown form of matter which only interacts gravitationally. Now, we can move on to the most common hypotheses for the true identity of dark matter.

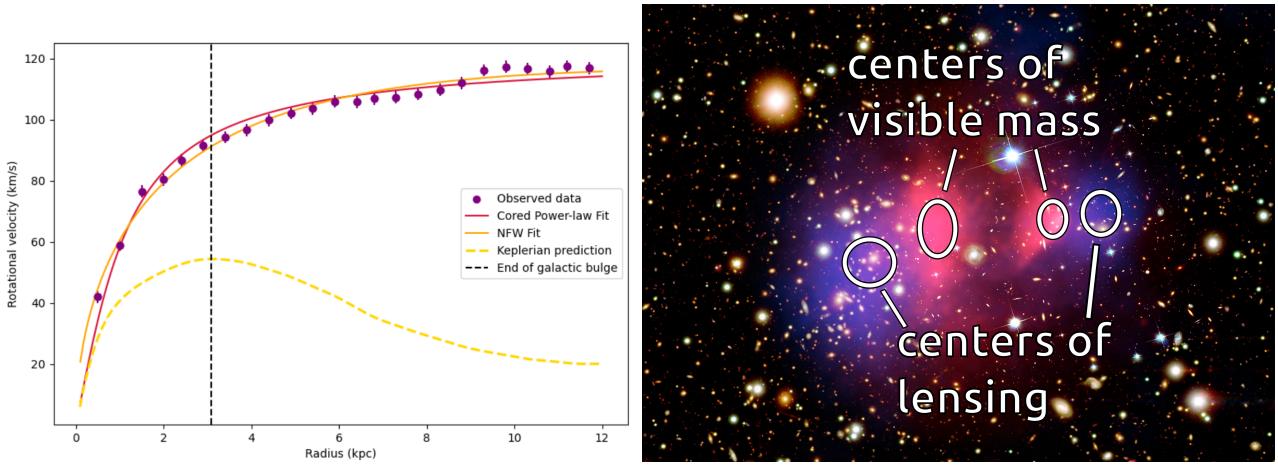


Figure 1: a: a plot of the predicted and observed data and various DM model fits for the galaxy M33. Made with matplotlib. b: ESA's Bullet Cluster photo, annotated by E. Ducluzeau to show offset lensing.

3 Theories Of Dark Matter

3.1 What About The Standard Model?

One final obvious question arises: why not look in capricious parts of the Standard Model (SM) for our dark matter particle? The SM does have some weird stuff. As it turns out, the SM does contain the neutrino, a particle of near-zero mass which very rarely interacts with other forms of matter and only through the weak force, making it invisible. It is detected only very rarely in specially designed neutrino detectors by observing the Cherenkov radiation left behind from a neutrino-particle collision.

Early universe particle populations eventually "freeze out" once the universe expands too much for them to stay in thermodynamic equilibrium, leading to a slow decline in their population modulated by their annihilation cross-section and officially marking their status as relics; this phenomenon is known as "decoupling". At sufficiently high pressures, usually occurring for relativistic particles where $T >> m$, particles mainly interact via inelastic scattering and remain in "thermal equilibrium" while the rate of scattering remains constant; "kinetic decoupling" is the point where this stops being true. At low pressures, usually where $T << m$, particles instead remain in "chemical equilibrium" where the rate of forward and backward particle production interactions is constant.

As they are ultrarelativistic, neutrinos decouple kinetically before decoupling chemically, and this occurs when $T_f \geq m_\nu$ at approximately 1MeV. Thus, the theory of neutrinos as DM is known as a "hot dark matter" theory. Knowing this, we can get the neutrino's abundance from thermal equilibrium at freeze out by solving the collisional Boltzmann equation[4], an equation governing the time-evolution of the number density of a given object allowing identification of when it decouples:

$$a^{-3}\frac{d(n_1)a^3}{dt} = \Pi_{i=1}^{4} \overbrace{\int \frac{d^3p_i}{(2\pi)^3 2E_i}}^{\text{Lorentz invariant phase space}} \overbrace{(2\pi)^4\delta^{(3)}(p_1+p_2-p_3-p_4)\delta(E_1+E_2-E_3-E_4)}^{\text{energy-momentum conservation}} |\mathcal{M}|^2 (\underbrace{f_3 f_4[1\pm f_1][1\pm f_2]}_{\text{backward scattering}} - \underbrace{f_1 f_2[1\pm f_3][1\pm f_4]}_{\text{forward scattering}}) \tag{5}$$

This equation is big and scary, but there is fortunately a less intimidating version that works for models at freeze-out[5]:

$$\frac{dY}{dx} = -\frac{xs(T)}{H(x)}\langle\sigma v\rangle \qquad\qquad H^2(x) = \frac{\pi^2}{90}g_{*,R}(x)\frac{x^4}{M_{\mathrm{Pl}^2}}(Y^2 - Y_{\mathrm{eq}}^2) \tag{6}$$

Further solving discussed in the citations obtains a concise equation for neutrino abundance in terms of x at freeze out:

$$Y_{\mathrm{eq}}^{\nu}(x_f) = n_{\mathrm{eq}}/s = 0.21g/g_{*s}(x_f) \tag{7}$$

By using the known g value of today (2) for fermions, the equation

$$\rho_{\nu 0} = m_\nu n_{\nu 0} \tag{8}$$

and the corresponding values of T and g_{*s} for both equilibrium and today, the relic abundance is obtained through the equation:

$$\Omega_\nu h^2 = \frac{\rho_{\nu 0}}{3M_h^2 M_{\mathrm{Pl}}^2} \approx \frac{\sum m_\nu}{100\mathrm{eV}} \tag{9}$$

We can match this against two crucial things, the known dark matter relic density, and the "Tremaine-Gunn bound". The dark matter relic density is obtained by using a rewritten Friedmann equation in terms of density parameters:

$$\Omega_c = 1 - \Omega_b - \Omega_k - \Omega_\Lambda \tag{10}$$

When measured via Big Bang nucleosynthesis, $\Omega_b h^2$ turns out to equal 0.022, with an Ω_Λ of 0.7 and an Ω_k of 0 due to the flatness of the universe. Performing the subtraction, the dark matter relic density $\Omega_c h^2$ is measured to be 0.12, which is the relic density used to compare all theories of DM.[6] Dark matter is also known from astrophysical and cosmological observations to cluster around galaxies in large spherical formations called "halos". The Tremaine-Gunn bound is the known minimum mass of $m_f = 120\mathrm{eV}$ for fermionic DM to form halos.[7] Putting all of this information together, we see a contradiction between Equation **??** and the Tremaine-Gunn bound; the neutrino mass would have to be 12eV to give the correct DM relic density, which would be in violation of the bound. As we are now out of SM candidates, we will have to look elsewhere.

3.2 WIMPs

So, what does that mean for particle dark matter? The most prominent particle-based theory of dark matter is that dark matter is the light, incredibly difficult to detect, and fittingly named WIMP (which stands for "weakly interacting massive particle"), or "cold dark matter" due to its low velocity. Despite the tall order that is motivating the existence of a particle beyond the SM, this can be done in two separate ways, in one case without explicitly trying to solve dark matter, which makes the WIMP theory attractive to scientists. The first way of doing this is to explicitly try to solve dark matter and somehow find the properties of a similar particle to the neutrino but that obeys the Tremaine-Gunn bound, which logically would probably be a really slow neutrino-like particle. Fortunately, we can use the collisional Boltzmann equation 6 for this as well. As the WIMP should be a nonrelativistic particle, we expect chemical decoupling to happen first. This assumption does a few key things: firstly, it means that the WIMP population stays in thermal equilibrium throughout this entire process. This leads to the idea that both x and $\frac{dY}{dx}$ tend to 0 at high temperatures. This paradigm also models a power law dependence for the interaction cross-section and an unchanging number of relativistic degrees of freedom. Putting all of this together, the simplified abundance equation for WIMPs is:

$$Y_\infty \approx \frac{(n+1)x_f^{n+1}}{\lambda} \tag{11}$$

By supplying the full equations for the number density and Hubble parameter for this nonrelativistic fermion at chemical equilibrium

$$n_{\text{eq}}(x) = \frac{g_\chi}{(2\pi)^{3/2}} m_\chi^3 x^{-3/2} e^{-x} \qquad H(x) = \frac{\pi}{3\sqrt{10}} \frac{m_\chi^2}{M_{\text{pl}}} g_*(x)^{1/2} x^{-2} \tag{12}$$

and plugging them into the abundance equation, a relation between the abundance and the mass of the WIMP is found:

$$Y_\infty \approx 0.4 \left(\frac{\text{eV}}{m_\chi} \right) \tag{13}$$

From there, different combinations of mass and cross-section can be tried to see which models of the WIMP's properties theroetically do and don't work.

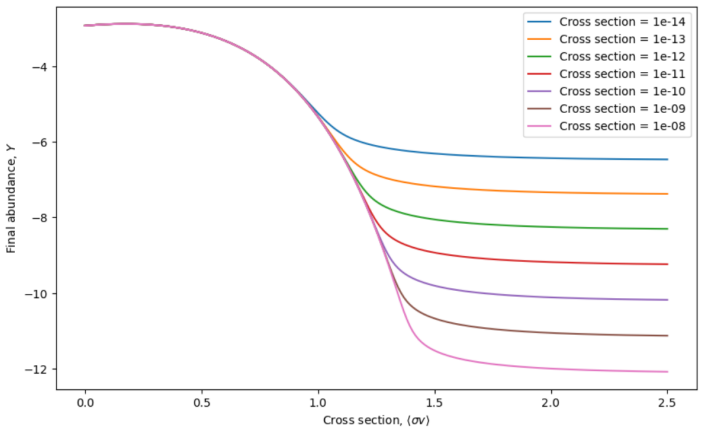


Figure 2: The relation between WIMP abundance and various starting cross section estimates assuming the WIMP mass to be 100GeV, plotted logarithmically. Made in Python

In an exciting turn of events, the theory of supersymmetry (SUSY) also accidentally predicts dark matter. SUSY is a popular theory of particle physics, where by implementing an extension of the Poincaré group which dictates the allowed particles and interactions of the SM, each spin-1/2 particle gains a corresponding spin-1 "superpartner" particle and vice-versa. One paradigm known as the "minimal supersymmetric Standard Model" is a SM extension which adds the minimum possible amount of supersymmetric particles (sparticles) to formulate a model satisfying the constraints of SUSY. This model was originally developed to solve the "hierarchy problem", which itself arises from "renormalisation" in quantum field theory (QFT), which is a phenomenon where the true value of a quantum property is offset from the measured value by quantum effects. In the case of the Higgs field, the quantum contribution to its measured mass-squared parameter is[8]:

$$\delta m_H^2 \sim \left(\frac{\alpha}{2\pi} \right) \Lambda^2 \tag{14}$$

Here α is the fine structure constant and Λ^2 is a dimensional scaling parameter, which we expect to be approximately $10^{18} GeV$ in a universe where quantum gravity holds. This means that the quantum contribution to the true mass-squared parameter is approximately 16 orders of magnitude greater than the measured value of $125.1 GeV$; this is the hierarchy problem. MSSM adds five parameters, each around the TeV scale, to form the SUSY Lagrangian[9]:

$$\mathcal{L} = \mathcal{L}_{\text{SUSY}} + \mathcal{L}_{\text{soft}} \tag{15}$$

where the "SUSY" Lagrangian contains all supersymmetric interactions and the "soft" Lagrangian contains positive mass and coupling parameters that reduce supersymmetry to observed non-supersymmetric physics. This changes the quantum contribution to the Higgs mass-squared parameter to

$$\delta m_H^2 = m_{\text{soft}}^2 \left[\frac{\lambda}{16\pi^2} \ln \left(\frac{\Lambda_{\text{UV}}}{m_{\text{soft}}} \right) + ... \right] \tag{16}$$

which notably no longer has a quadratic dependence on a singular dimensional scaling parameter, resolving the hierarchy problem. Under this model, the lightest sparticle is the only stable sparticle, to which all other sparticles decay if not to SM particles, explaining how it could have formed into a relic as per the theory of WIMPs[10]. Taking known bounds of the MSSM model that solves the hierarchy problem, the lightest neutralino (neutron superpartner) has a mass of approximately 100GeV and couples to the Z on the scale of the weak, which when plugged into Equation 13 returns an interaction cross-section of $10^{-26} cm^3 s^{-1}$, which yields a neutralino relic density of 0.12, the exact amount predicted to be required through Equation 10; this supersymmetric coincidence is hailed as the "WIMP miracle". In trying to solve a different problem by invoking supersymmetry, we accidentally solve dark matter. Nice.

3.3 Axions

The most prominent wave-based theory of dark matter is the axion. Once again, axions in the context of DM arise accidentally, in this case through trying to solve a separate problem in quantum chromodynamics (QCD). Axions were first formulated to solve the strong CP problem, which is the curious observation that even though QCD allows for charge-parity violations this is never observed in strong-only interactions. In QCD, the Lagrangian for a theory containing just a quark and nothing else would be:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \theta \frac{g^2}{32\pi^2} F_{\mu\nu} \tilde{F}^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m e^{i\theta' \gamma_5})\psi \tag{17}$$

Here the second term is the vacuum angle and the fourth term is the quark mass term, both capable of violating CP symmetry where $\theta' \neq 0$. Converting this principle to the Standard Model, the vacuum angle becomes $\bar{\theta} = \theta - \text{argdet}(Y_u Y_d)$ where $\bar{\theta}$ has been shown to be smaller than the nanoscale despite being able to take any value between 0 and 2π.

One possible resolution to this is to introduce a new complex scalar field Φ into the particle physics blender described by

$$\mathcal{L} = -\partial_\mu \Phi^* \partial^\mu \Phi - \lambda \left(|\Phi|^2 - \frac{v^2}{2} \right)^2 \tag{18}$$

which we assume couples to the early universe thermal bath.
While the temperature is very high, the field is kept from spontaneously breaking its own symmetry (SSB), however once the temperature gets low enough SSB is possible, the field becomes $\Phi = \frac{v e^{i\theta}}{\sqrt{2}}$, and the Lagrangian becomes:

$$\mathcal{L} = -\frac{1}{2} v^2 \partial_\mu \theta \partial^\mu \theta \tag{19}$$

Here, θ is the Goldstone boson (bosons arising from SSB) for this complex scalar field, and rewriting this 0 dimensional parameter as a 1 dimensional parameter $\phi = v\theta$; this ϕ is the "axion". Once the axion is defined, several theories such as the Peccei-Quinn model can extend it to a solution of the strong CP problem; for example, the Peccei-Quinn model introduces a quark-like parameter Q described by the Dirac equation in addition to the axion, which after some solving leads to the Lagrangian:

$$\mathcal{L} = \frac{1}{2}(\partial\phi)^2 + \frac{\partial_\mu \phi}{2 v_a} \bar{Q} \gamma^\mu \gamma_5 Q + \frac{\phi}{f_a} \frac{g_3^2}{32\pi^2} G\bar{G} \tag{20}$$

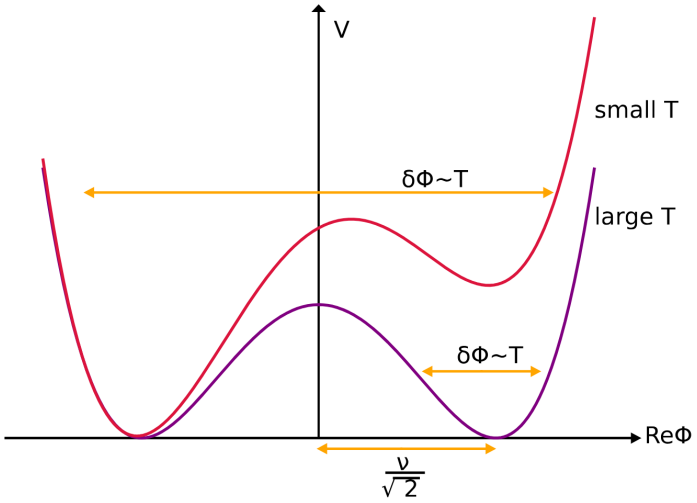


Figure 3: Plot showing how small temperature fluctuations are required for the true minimum to be accessible for SSB, and how temperature affects the axion potential. Made in Python

Notably, the parameter $\bar{\theta}$ is gone, and plugging this equation into the Klein-Gordon equation tends to 0, solving the strong CP problem[11]. This is important because it shows that the axion has merit outside of dark matter, but the fact that this scalar field carries mass-energy means that it turns out to also be a strong candidate for DM. As previously mentioned when talking about WIMPs, we like candidates for DM that arise naturally as a result of trying to do something else.

As the axion is a scalar field, its properties are obtained by solving the Klein-Gordon equation for an arbitrary spacetime:

$$\Box\phi - m^2\phi = 0 \tag{21}$$

For the axion, this turns out to be a harmonic oscillator damped by the Hubble expansion. For the radiation-dominated era under the Wentzel-Kramers-Brillouin approximation, the solution

$$\phi \approx \phi_i \left(\frac{a}{a_{\text{osc}}}\right)^{-\frac{3}{2}} \cos\left(3H(t_{\text{osc}})t\right) \tag{22}$$

is obtained, which when combined with the WKB axion density equation

$$\rho_\phi \approx \frac{1}{2}m^2\phi_i^2(\frac{a}{a_{osc}})^{-3} \tag{23}$$

yields an equation for the axion relic density:

$$\Omega_a h^2 = 0.12 \left(\frac{\phi_i}{1.5\times 10^{15}\text{GeV}}\right)^2 \left(\frac{m}{10^{-15}\text{GeV}}\right)^{\frac{1}{2}} \tag{24}$$

Notably, very small masses and very large initial field values are required to obtain the right value for the axion relic density. Depending on what specific model of the axion is used, this may or may not be explainable.[12]

3.4 Primordial Black Holes

The last popular candidate for dark matter is also the most unusual one, focusing on very macroscopic and incredibly heavy objects rather than the unseen light abundances that come with WIMP and axion theories. This object is the primordial black hole (PBH), which is a black hole that would have somehow

formed very early in the universe, before there were stars to collapse! Despite the boldness of this claim, the theory remains attractive because it solves quite a few problems in space physics, such as potentially being "seeds" for the later emergence of supermassive black holes[13], or the new findings in physics that could emerge on the way as we search for PBHes. Additionally, we already know that black holes exist, so it is tempting to model DM as an extension of an existing phenomenon. The theory is also compatible with the restriction on early universe baryon counts derived from the light element abundances, because PBH contributor particles would causally disconnect from the universe and therefore not contribute to the light element abundances.

The only issue with the PBH theory is that they would have had to form well before recombination, wherein everything was still a plasma of baryons and photons so large-scale structures could not form. This means that they'd have to form by some exotic mechanism which could collapse matter into a PBH before there were stars. Fortunately, as the early universe plasma was an unstable fluctuating structure, it could provide such a mechanism. As a theory of spacetime distortion, we can solve the Friedmann equation for a patch of early universe spacetime undergoing mass-energy-induced curvature fluctuations to derive a PBH model. Modelling such a perturbation ζ as the spherical spacetime

$$ds^2 = -dt^2 + a^2\left[\frac{dR^2}{1-K(R)R^2} + R^2 d\Omega^2\right] \tag{25}$$

where $R = re^{\zeta(r)}$ and $K(R) = -\frac{\zeta'(r)}{r}[\frac{2+r\zeta'}{e^{2\zeta}}]$, we want to find the conditions where such a perturbation would accumulate enough mass-energy to collapse into a PBH. It turns out that this spacetime resembles a closed universe with curvature K, allowing application of the closed-universe Friedmann equation:

$$H^2 + \frac{K}{a^2} = \frac{8\pi G}{3}\rho \tag{26}$$

The part of this equation describing the perturbation is the part in K, which can be labelled as:

$$\delta = \frac{K}{H^2 a^2} \tag{27}$$

We also know that for a closed universe, turnaround happens when the scale factor reaches an extremum; at this point, the closed universe approximation no longer applies and the perturbation reconnects with the rest of the universe, forming a black hole if the curvature is perturbed more than

$$\delta > \delta_c = c_s^2 = \frac{1}{3} \tag{28}$$

which can then be used to define the collapse fraction β (the fraction of the universe where $\delta > \delta_c$) and finally a way to find the PBH relic density:

$$\Omega_{PBH}h^2 = 0.12\left(\frac{\Omega_m}{0.3}\right)\left(\frac{h}{0.7}\right)^2\left(\frac{1+z_{\text{eq}}}{3400}\right)^{-1}\left(\frac{T_{\text{collapse}}}{1\text{GeV}}\right)\left(\frac{2.725\text{K}}{T_0}\right)\left(\frac{\beta}{6.5\times10^{-10}}\right) \tag{29}$$

Getting the right relic abundance requires fixing β to a specific very small value, which is found through the PBH mass function[14]:

$$\beta(M_H) = 2\int_{\delta_c}^{\infty}\frac{M_{\text{PBH}}}{M_H}P(\delta(R))d\delta(R) \tag{30}$$

This provides a plausible mechanism by which PBHes could form in the early universe before there were stars. Whether or not PBHes can make up all the DM is determined by the range of masses of PBHes that formed in the early universe. The final piece of the PBH puzzle is piecing together the conditions for such curvature perturbations to arise in the first place; one such model is a slow-roll approximation for universal expansion, which can yield perturbations of around $\zeta \sim 0.1$ at the edge of the cosmic event horizon if the slow-roll has exactly the right shape. When the universe shifts to radiation domination, these perturbations can grow large enough to achieve $\delta > \delta_c$ and therefore form PBHes.[15]

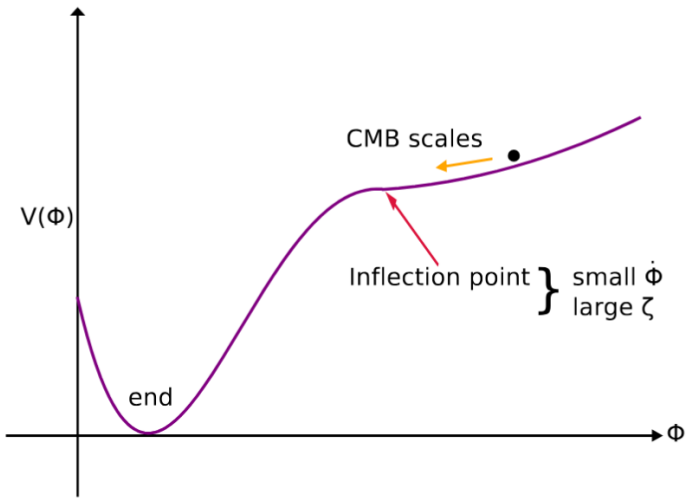


Figure 4: Slow roll model for PBH generation; at recombination, the inflection point is passed and ζ cascades in size. Made in Python

4 Phenomenology In The Lab And Beyond

Each theory of dark matter has its own consequences for space physics and implication in methods of detection. We will look at these in order of discussion.

4.1 WIMPs

The WIMP has the main interaction routes typical of particles, with direct detectors looking for WIMP-induced recoil of a SM particle or the production of a WIMP in a collider, while telescope experiments attempt to indirectly detect the photonic fallout of WIMP self-annihilation.[16] To directly detect WIMPs, we would need to estimate the energy of a nucleus recoiling from WIMP impact and the WIMP cross-section. The WIMP impact equations for 4-momentum transfer and nuclear recoil energy are given as:

$$q_\mu = (E_R, \sqrt{2m_N E_R}\cos\theta, \sqrt{2m_N E_R}\sin\theta, 0) \tag{31}$$

$$E_R = \frac{(m_N m_\chi)^2}{(m_N + m_\chi)^2}\frac{v_i^2}{m_N} \tag{32}$$

For a WIMP-nucleon interaction, the cross-section for the whole nucleus is derived from the matrix element of quantum interactions between DM, protons, and neutrons to be:

$$\theta_N = \frac{1}{m_{Z\prime}^4}\left[g_\chi\left(2g_u + g_d\right)\right]^2 \mu_p^2 \left(\frac{\mu_N}{\mu_P}\right)\frac{\left[f_p Z + f_n(A-Z)\right]^2}{f_p^2} \tag{33}$$

Finally, these can be taken together to give the interaction rate:

$$R = \frac{\rho_{DM}}{m_\chi}\langle\sigma\nu\rangle \tag{34}$$

This is used by direct detectors to get the total estimated WIMP interactions for some parameters of WIMP mass, cross section, and detector runtime. Finally, a rarer but theoretically possible type of direct DM detection would be to produce a WIMP in a collider, such as by SUSY decays.

We also think that WIMPs could self-annihilate and release observable gamma rays, allowing us to

indirectly detect DM. The Boltzmann equation for the NFW DM profile provides a starting point for this:

$$\frac{dn_\gamma}{dt} = \sum_i \frac{n_\chi^2}{2} \langle \sigma_i \nu \rangle N_i \tag{35}$$

This leads to an equation for flux $d\Phi$ in a given energy band dE_γ in the direction ψ:

$$\frac{d\Phi}{dE_\gamma} = \frac{A}{4\pi} \left[\int d\ell \rho^2 \left(r \left(\ell, \nu \right) \right) \right] \frac{1}{2m_\chi^2} \sum_i \langle \sigma_i \nu \rangle \frac{dN_i}{dE_\gamma} \tag{36}$$

The gamma ray flux we detect is then suppressed due to propagating through the galaxy:

$$\frac{dN}{dt} = \overbrace{D\nabla^2 N}^{\text{diffusion}} + \overbrace{\frac{\partial}{\partial E}(Nb)}^{\text{loss}} + \overbrace{Q}^{\text{injection}} \tag{37}$$

This shows that, owing to the predicted rarity of DM self-annihilation and the suppression of the gamma flux, this requires areas that are both dense in DM and low in dust to do indirect detections, such as dwarf galaxies.[17] Challenges exist to each approach. The neutrino fog, which is the effect resulting

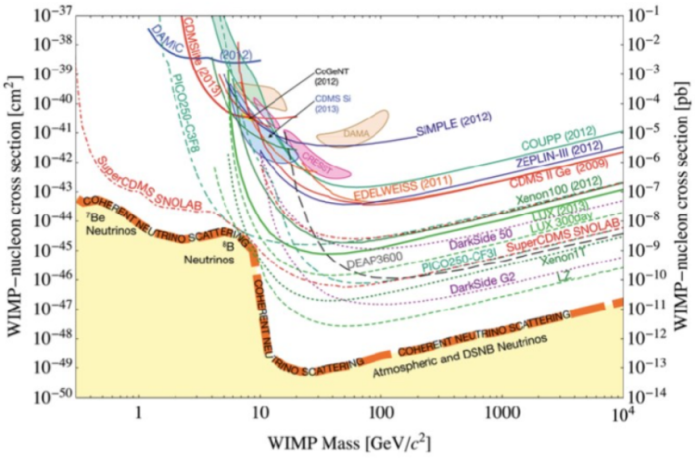


Figure 5: Constraints on the WIMP mass and cross section as a result of all the experiments.[18]

from the fact that neutrinos also pass through DM detectors which will eventually become big enough to detect them despite their small cross section, would require new techniques to distinguish between neutrino and DM signals once reached. Indirect detection is also challenged by the fact that DM annihilation isn't the only source of gamma ray bursts with other candidates such as millisecond pulsars being theorised, also requiring further refinement to be able to distinguish these sources.

Thus far, experiments have included direct detectors such as LUX-ZEPLIN and CRESST which using things such as large tanks of dense and highly scintillant noble gas or supercooled metallic disks respectively to detecting aftermaths such as photons or phonons from a WIMP interaction, and indirect detectors such as the MAGIC and FLAT telescopes which aim to detect gamma rays via cascades on Earth or directly in space respectively. As of yet, the WIMP has not been detected, but running all of these experiments has put strict bounds on the WIMP's mass and cross section. In recent times, the lack of evidence for WIMPs and SUSY has led to a slight decline in interest in the theory[19] compared to alternatives like axions, but the search still continues.

4.2 Axions

As a scalar field, the axion ϕ is detected according to the rules of QFT. It's theorised to couple to the electromagnetic field, observable by detecting a field oscillating according to

$$\nu = \frac{m_a}{2\pi} \approx \frac{1Hz}{2\pi}\frac{m_a}{10^{-15}\text{eV}} \tag{38}$$

which couples as

$$g_{\text{QCD}} = 10^{-15}\left(\frac{m_a}{5\times 10^{-6}\text{eV}}\right)\text{GeV}^{-1}C_{\phi\gamma} \tag{39}$$

resulting in an axion-photon Lagrangian:

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m_a^2\phi^2 - \frac{g}{4}\phi F_{\mu\nu}\tilde{F}^{\mu\nu} \tag{40}$$

When this is plugged into Maxwell's equations, the axion field interaction modifies them such that electromagnetic induction acts in parallel rather than perpendicular, a key avenue for axion detection.[20]

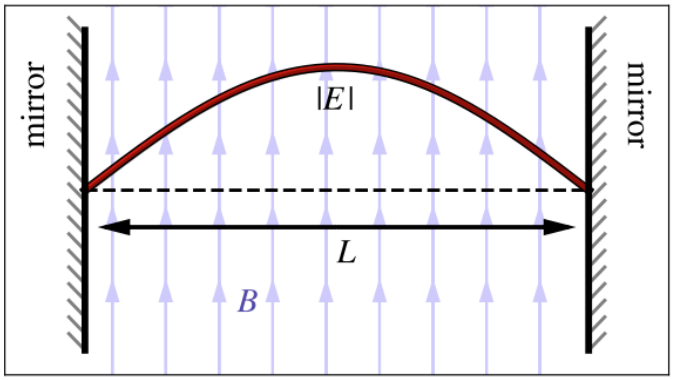


Figure 6: Diagram to show what a haloscope cavity is.[clastex]

The most common axion detector is the haloscope, which consists of a cavity placed in a strong magnetic field which aims to detect photons produced by the axionic interaction with this field[21]. In the haloscope, the induced electric field from this magnetic field acts as a driven harmonic oscillator:

$$\ddot{E} - \nabla^2\vec{E} = \frac{\omega^2}{m}\sqrt{2\rho_a}g\vec{B}\cos(\omega t) \tag{41}$$

For the modes $E_n \propto sin\left(\frac{n\pi x}{L}\right)$, resonance is achieved at $\frac{n\pi}{L} = \omega$, which is damped by physical factors as:

$$Q = \frac{\overbrace{\omega}^{\text{energy stored}}}{\underbrace{\Gamma}_{\text{energy loss}}} \tag{42}$$

where Γ is the "loss factor". The induced power that leaks out of the cavity can be observed, which is what allows haloscopes to theoretically detect axions:

$$P = \kappa C\frac{g^2B^2V\rho_{\text{DM}}}{m}\min\left[Q, \sigma_\nu^{-2}\right] \tag{43}$$

As the axion mass isn't known, we don't know what the resonant frequency is either, meaning that a good axion detector needs to be able to scan ranges of frequencies, with a shift in the scanned resonant frequency modulated by:

$$\Delta\nu \sim \frac{\nu_{\text{res}}}{Q} \tag{44}$$

Axion detectors also have signal-to-noise ratios given by:

$$\text{SNR} = \frac{P_s}{T_{textsys}\sqrt{\Delta\nu}}\sqrt{\Delta t} \tag{45}$$

Ideally, an axion detector should have an SNR of 3.[4]

The ADMX experiment is one such detector, which uses two tuning rods to change its effective cavity length and therefore scanned resonant frequency. ADMX searches for photons, but the axion may also couple to other things, such as nucleons or electrons which the QUAX experiment aims to detect. The search for axions has experienced an uptick in recent years, motivated by WIMP searches turning up empty and the allure of axion applications to other relevant problems such as the strong CP problem. Axions are becoming increasingly well-bounded from the world's experiments, with a sweet spot in the microwave range next in line for investigation.[22]

4.3 Primordial Black Holes

Black holes are widely thought to emit radiation in the form of "Hawking radiation", which occurs due to quantum fluctuations on curved spacetime. Usually, a spontaneously produced matter-antimatter pair self-annihilates or falls completely into the black hole, but occasionally the black hole absorbs only one of the particles; in order to conserve stress-energy, the black hole must consequently lose mass while the other particle is allowed to fly freely as radiation. Eventually, a black hole with no mass to accrete will stop existing[23], and a lifetime formula can be found:

$$\tau_{\text{BH}} \approx \frac{2.7\times10^{14}}{\langle f(M)\rangle}\left(\frac{M}{10^{14}\text{g}}\right)^3 s \tag{46}$$

For PBHes to be DM, they need to live longer than the current age of the universe, which leads to the mass bound of $M_{\text{PBH}} \gtrsim 2.5\times10^{-19}M_\odot$ from the lifetime equation. If all DM was PBHes at this mass bound, we expect 10^{-7} detections a year for objects on the femtometer scale of length, and this only goes down as the black holes get heavier. What this means is we're probably never going to directly detect one, and we also don't want to, because this would cause devastating earthquakes or worse.

Instead, we try to constrain them so that we can hopefully indirectly detect them via microlensing. The emission of Hawking radiation gives black holes a temperature:

$$T_{\text{BH}} = \frac{1}{8\pi GM} \tag{47}$$

Using all of these equations, upper and lower bounds on black hole parameters can be set. The lower mass bound is given by observations of the universe's high frequency photons, as we know that the Hawking radiation emission peaks at:

$$I(T_{\text{PBH}}) \propto T_{\text{PBH}}^2 f(M)\Omega_{\text{PBH}} \tag{48}$$

As this can't exceed the observed spectrum, the black hole mass can't get too small as this would shift the spectrum too high, leading to the lower bound of $M_{\text{PBH}} \gtrsim 10^{-16}M_\odot$. The upper mass bound, meanwhile, is motivated by the survival of fragile structures such as Eridanus II, which is only possible if its

dark matter is light enough to not disturb it[24]; for a PBH, this is bounded at $M_{\text{PBH}} \lesssim 3M_{\odot}$.

With these bounds in place, gravitational microlensing can be used to search for PBHes, which is a sub-phenomenon of gravitational lensing where the lensing object is too small to cause the usual lensing effects, rather increasing the lensed object's apparent magnitude. The change in magnitude for microlensing as arranged on a 2D plane is given by:

$$\mu_{\text{tot}} = \frac{y^2+2}{y\sqrt{y^2+4}} \tag{49}$$

Detectors survey given stars at intervals of Δt, after which any magnification with $\mu > 1.34$ and a light curve shape close enough to a known template being flagged as a microlensing event. For microlensing events with time:

$$\hat{t} \approx 10^7 \left(\frac{M}{M_{\odot}}\right)^{\frac{1}{2}} \left(\frac{D_s}{50\text{kpc}}\right)^{\frac{1}{2}} \tag{50}$$

A microlensing search expects to observe an amount of events given by:

$$N_{\text{exp}} = E \int_0^{\infty} \frac{d\Gamma}{d\hat{t}} \epsilon(\hat{t}) d\hat{t} \tag{51}$$

where E is the number of stars observed multiplied by runtime, $\epsilon(\hat{t})$ is the observation efficiency, and $\frac{d\Gamma}{d\hat{t}}$ is the expected rate of microlensing events.
Experiments involve looking for microlensing with telescopes. Surveys such as the EROS[25] and MACHO surveys have attempted this, choosing to look for microlensing events with long times in the Milky Way. The Subaru-HSC[26] search meanwhile looked for microlensing events with short times in Andromeda. Due to none of these observing microlensing events, the remaining viable PBH masses to be DM are constrained by the results of the experiment. Following the decline in popularity of WIMPs and advancements in space science such as LIGO and JWST, PBH theory is seeing increased attention, but owing to the difficult nature of detection, many remain skeptical.

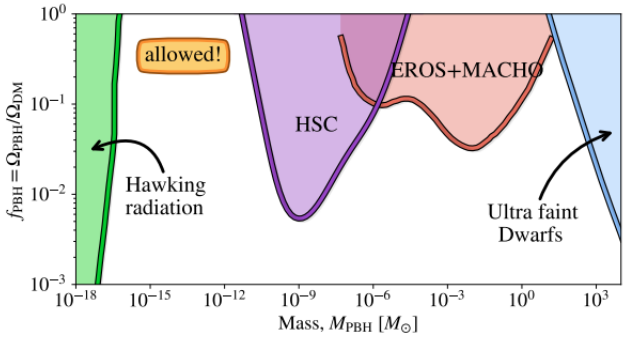


Figure 7: Constraints on the PBH mass as a result of all the experiments.[4]

5 A Brief Look Into Alternative Theories

Modified Newtonian dynamics (MOND) is a proposal that, rather than invisible flying stuff, the gravitational anomalies are simply explained by our theory of gravity being wrong. In 1983, the Israeli Mordehai Milgrom proposed a modification to gravity on the largest scales by changing how the effective acceleration term in $F = ma_F$ works, which would affect the orbits of celestial bodies while leaving its effects at human scales untouched, as follows[27]:

$$a_F = \begin{cases} \frac{GMm}{r^2} & \text{for } a >> a_0 \\ \\ \frac{\sqrt{a_0 GM}}{r} & \text{for } a << a_0 \end{cases} \tag{52}$$

where a_0 is some threshold for the Newtonian acceleration.

MOND is able to explain certain anomalies in galactic dynamics which ΛCDM fails to explain, however it is unable to explain anomalies not directly tied to visible mass such as the Bullet Cluster or CMB anisotropies; thus the theory has not gained widespread traction. Finally, as a nonrelativistic theory, relativistic corrections would be required before it yields any workable results.

This led to the development of tensor-vector-scalar gravity (TeVeS), a relativistic modification of MOND introduced in 2004 by Jacob Bekenstein, reformulating it as an action principle theory to ensure relativistic agreement and the validity of conservation laws from the start[28]:

$$\begin{gathered} S_{TeVeS} = \int \overbrace{(-\tfrac{1}{16\pi G} R\sqrt{-g}}^{\text{Einstein-Hilbert Lagrangian}} \\ \overbrace{-\tfrac{1}{2}[\sigma^2 h^{\alpha\beta}\delta_\alpha\phi\delta_\beta\phi + \tfrac{1}{2}\tfrac{G}{l^2}\sigma^4 F(kG\sigma^2)]\sqrt{-g}}^{\text{scalar Lagrangian}} \\ \underbrace{-\tfrac{K}{32\pi G}[g^{\alpha\beta}g^{\mu\nu}(B_{\alpha\mu}B_{\beta\nu}) + 2\tfrac{\lambda}{K}(g^{\mu\nu}u_\mu u_\nu - 1)]\sqrt{-g})}_{\text{vector Lagrangian}} \end{gathered} \tag{53}$$

This theory reproduces the flat rotation curves that MOND was originally designed to do, and the issues of stars with lifetimes on the order of weeks and gravitational lensing may be solved by the theory increasing the bounds on the neutrino mass. However, TeVeS is also notably unable to explain the Bullet Cluster.

Constantinos Skordis and Tom Złośnik in 2021 made a further attempt to formulate MOND relativistically while addressing the Bullet Cluster and CMB problems[29]. However, this attempt results in additional cosmological degrees of freedom; as shown by the theory of axions, these additional degrees of freedom could be interpreted as a form of dark matter, meaning at least some DM is still required.

6 A Brief Look Into Dark Matter Subcategories

6.1 Sterile Neutrinos

In between the "cold" DM theory of WIMPs and the "hot" DM theory of "active" neutrinos is the "warm" DM candidate, the "sterile neutrino". There exists a mass limit in between active neutrinos and WIMPs that a hypothetical particle with in between mass and velocity could fill while still being dark matter, and such a particle is named the sterile neutrino. Its existence is motivated by an anomaly in lepton chirality where active neutrinos are the only lepton observed to only take one of the two handedness values. They're additionally motivated by the fact that the Standard Model doesn't predict active neutrino masses, which sterile neutrinos with a mass M_{NHL} could solve through the seesaw matrix:

$$M_v \sim \begin{pmatrix} 0 & m_D \\ m_D & M_{\text{NHL}} \end{pmatrix} \qquad \lambda_\pm = \frac{M_{\text{NHL}} \pm \sqrt{M_{\text{NHL}}^2 + 4m_D^2}}{2} \tag{54}$$

The sterile neutrino is predicted to be very heavy, and the m_D mixing terms theoretically allow for the sterile neutrino mass to "leak" into the active neutrino, giving it mass.[30]

This is another model of DM which is attractive because it solves another problem and accidentally predicts DM in the process; sterile neutrinos, being heavy massive particles with no hypercharge not traveling at ultrarelativistic speeds, are a very viable DM candidate. However, this also means that observing one would pose a great challenge, with prior experiments to investigate SUSY constrain the sterile neutrino mass to the hard-to-detect keV scale. As time passes and WIMP searches come up short, the theory of sterile neutrinos is gaining popularity[sterile2], however the sheer difficulty of detection means that other candidates are likely to be pursued beforehand. Since they're probably relativistic, they also can't be all the DM by themselves.

6.2 WIMPZILLAs

"WIMPZILLA" theory is a theory of heavy dark matter which arises from considering the possibility of non-thermal production of WIMP-class particles in the early universe, posited by Edward W. Kolb, Daniel J. H. Chung, and Antonio Riotto in 1998. A WIMPZILLA model assumes that the rate of expansion is faster than the rate of annihilation of DM particles, meaning that the WIMPZILLAs aren't in thermal equilibrium by the time of chemical decoupling, with this non-equilibrium requirement leading to an equation for the WIMPZILLA mass:

$$\left(\frac{200\text{TeV}}{M_\chi}\right)^2 \left(\frac{T_*}{M_\chi}\right) < 1 \tag{55}$$

This allows some truly ludicrous WIMPZILLA masses, potentially ranging all the way to the PeV scale.[31] If the WIMPZILLA is the answer to dark matter, detection of the particle would focus on measuring the universe's expansion rate at the end of inflation, which would allow bounds to be put on the WIMPZILLA and for a proper search to begin, as the WIMPZILLA could have parameters such as electric charge that the regular WIMP does not have. However, the theory relies on supersymmetric theoretical elements such as gravitinos to determine some of its bounds, and so has not been widely accepted as a real candidate worth investigating.

6.3 Macroscopic Dark Matter

Macroscopic dark matter is similar to the PBH theory, but rather than invoking black holes, it aims to explain DM by postulating the existence of clusters of macroscopic dark matter which could be observed with the naked eye... if they were visible.

Two running theories exist. The first is massive compact halo objects (MACHOs), which postulates that low- to no-emission objects float around galaxies uninterrupted to cause the observed gravitational anomalies. Anything sufficiently heavy could be a MACHO, such as a neutron star, a brown dwarf, or a black hole with no nearby matter to form a visible accretion disk [macho].

The second theory is simply known as "macros", which are objects made of macroscopic clusters of particles, be they Standard Model or otherwise. The mass limit of macros for a formation T_* has been formulated to be:

$$M_{H,\text{dark}} \sim 10^{35}\text{g}\left(\frac{10^9\text{K}}{T_*}\right)^3 \tag{56}$$

Theories thus far range from dense clusters of quark-gluon plasma, strangelets, or even an unknown type of dark matter particle; the low observation rate is explained by the fact that there are simply very few macros in space at any given time, so their probability of intersecting any human detection attempt becomes astronomically low.[32]

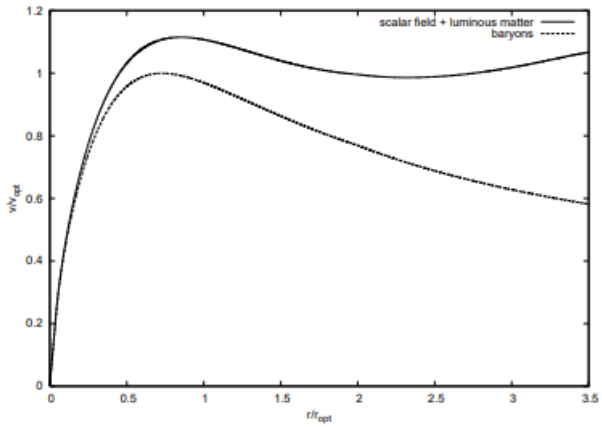


Figure 8: A comparison between the Keplerian and dark fluid predictions for rotation curves.

6.4 Dark Fluid

In 2005, cosmologist Alexandre Arbey suggested "dark fluid", a wave-type framework of dark cosmology which attempts to explain both dark matter and dark energy as a single phenomenon, which behaves like dark matter on galactic scales and like dark energy at even greater scales than that. The behaviour of dark fluid is given by a Lagrangian with three distinct parameters to determine its behaviour at galactic, cluster, and cosmic scales with a gauge that is invariant under sinusoidal oscillation:

$$\mathcal{L} = g^{\mu\nu}\partial_\mu\phi^\dagger\partial_n u\phi - V(\phi) \tag{57}$$

The potential now reads as:

$$V(\phi) = m^2\phi^\dagger\phi + \alpha\,\exp(-\beta\phi^\dagger\phi) \tag{58}$$

We can further constrain the three parameters m, α, and β to give the desired behaviours at the given scales:

$$l \sim \sqrt{\frac{1}{G^{\frac{1}{2}}\sigma_0}}\frac{\hbar}{mc} \qquad \alpha = \rho^0_{\mathrm{DE}} \qquad \beta = \sqrt{\frac{2\lambda}{\alpha}} \tag{59}$$

This returns the values $m \sim 10^{-23}\mathrm{eV}$, $\alpha \sim 7\times10^{-27}\mathrm{kgm}^{-3}$, and $\beta \sim 10^{-39}\mathrm{eV}^{-2}$. While the full math is difficult to work with, dark fluid naturally reproduces many candidates for both dark matter and dark energy, such as the galactic rotation curves or inflation, but also exotic theories like TeVeS or quintessence, which is also beyond the scope of this paper.[33]

7 Conclusion

This essay opened with the title question, "what could this be", and after examining all these candidates, we can finally answer the question with "it could be many things". There are ways to model dark matter as being something based in particles, waves, a combination thereof, or something else entirely, each with their own criteria and challenges in the detection effort. We continue to constrain but have yet to definitively prove any of them. Not all theories of dark matter are perceived equally; some get more funding and public attention, others get less, some are more difficult to test. Some even currently can't be tested, some don't even have constraints right now, but they all remain well formulated and worth exploring. We'd like the true identity of dark matter to be simple, but there's no law of physics saying it has to be. All that's left to do is to keep searching until we find out what it really is.

Acknowledgements

- Ethan Jones: mathematical assistance for the galaxy rotation graph
- Amy McCullough: computational assistance for the sombrero potential and slow-roll collapse graphs
- Kunal Mehta: mathematical assistance for the WIMP cross-section relation graph

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